DMA

AXI4 / Shared External Memory

Banked Buffers (IFM / Weights / OFM)

Data Streamer

Vector DMA Read / Write

Feature Input

Weight Input

Bias / Multiplier / Shift

Feature Output Row Buffer

AXI4-Lite MMIO

TriHaRiscv CPU

IRQ / Status

Conv. Core

Layer Controller

Decode

Dense / DW

Auxiliary

Auxiliary Operators

Add / Sigmoid

Pool / Resize → OFM

Transpose / Reshape

Conv. Datapath (Std / DW / PW)

MAC MAC ⋯ MAC MAC

MAC MAC ⋯ MAC MAC

Configurable Dense PE Array (M × 16)

MAC MAC ⋯ MAC MAC

MAC MAC ⋯ MAC MAC

DW16 — Channel-wise 3 × 3 Conv.

Add / Sigmoid quantization

Shared Q

Bias

Act.

Requant.